

Sabanci University

Spring 2025–2026

CS 310 – Mobile Application Development

Course Information

Level of Course	Undergraduate
Prerequisites	CS 204
Course Page	https://sucourse.sabanciuniv.edu/plus/course/view.php?id=12841

Time & Place

Time	Week Day	Place
09:40–10:30	Tuesday	FASS-G062
14:40–16:30	Wednesday	FASS-G062

Instructor Information

Mohammad Yusaf Azimi

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Teaching Assistant Information

To be announced.

Course Description

The objective of this course is to provide students with the skills needed to build mobile applications. The course starts with an introduction to the required programming environments and moves forward with creating stateful web services and developing mobile applications consuming web services. Upon successful completion of this course, students are expected to design, code and implement applications on mobile and hand-held devices with limited resources; understand web services; manage messaging with HTTP; and deploy/consume web services.

Course Learning Outcomes

1. Gain knowledge on Object-Oriented Programming techniques for Mobile and Web Applications.
2. Gain knowledge on developing multi-tier applications.
3. Get experienced on developing web APIs.
4. Create user interfaces and navigation for mobile applications.
5. Send and receive mobile application data via networking.
6. Gain general knowledge and experience on Mobile Applications, tools and techniques.

Course Objective

The main objective of this course is to provide students with the tools and skills needed to build applications for mobile platforms. The course starts with a brief introduction to Object Oriented Programming and moves forward with creating stateful web services and developing mobile applications consuming web services via the mobile platform of concern.

Course Materials

Resources

- All course materials will be available on SUCourse, including:
 - Lecture Notes
 - Example Codes
 - Project Steps

Technology Requirements

- Flutter: <https://docs.flutter.dev/get-started/install>
- DartPad: <https://dartpad.dev/>
- Dart Documentation: <https://dart.dev/guides>

Assessment

Assessment Method	Percentage (%)
Final	35%
Midterm	30%
Quizzes	10%
Group Project	25%

Notes

- **Midterm Exam:** Date to be determined
- **Final Exam:** Scheduled by Student Resources
- **Project:** 4–5 steps plus a final demo
- **Quizzes:** Best 2 out of 3 quizzes count toward the final grade

Requirements

- Final exam grade below **30** → **Fail (F)**, regardless of total grade.
- Missing all quizzes, midterm, and final → Grade **NA**.
- Missing midterm or final without approved excuse → Grade **F**.

Policies

Time Conflicts & Attendance

- Time conflicts will be approved only with the understanding that students take full responsibility for any missed lectures and any missed quizzes.
- Attendance is not mandatory but strongly recommended for effective learning and achieving a good grade.
- There will be 3 quizzes; the best 2 contribute to 10% of the total grade.

Make-Up

- No makeups for quizzes or project steps.
- If you miss a midterm or the final exam (you may not miss both) due to a documented valid excuse (e.g., a medical report), you will be permitted to take a make-up exam after the final exam period. If you miss both the midterm and the final, the make-up exam will replace only the final exam grade, and the midterm grade will remain zero.
- Makeup exams may be oral or written and are often more comprehensive.

Late Submission

Late submissions will be accepted for up to two days, with a penalty of **10 points per late day**. Submissions made within the first hour after the deadline will receive a reduced penalty of **5 points**.

Academic Integrity

- Sharing work or submitting similar solutions (with other students or AI) is considered plagiarism.
- Plagiarism cases will be referred to the Dean's Office for disciplinary action.

Special Notes

- Students are responsible for announcements made in lectures, SUCourse+, and email.
- Regularly check your Sabancı University email.
- Failure to follow announcements is not a valid excuse.

Use of AI

- You may use ChatGPT or similar tools to obtain limited programming assistance for your project.
- Any LLM usage must be for partial help only, not for generating complete answers.
- Submissions found to rely excessively on LLM-generated content may be assigned a grade of zero, and/or the student may be required to complete an oral examination.
- The use of LLMs or any AI tools during quizzes or exams is strictly prohibited. Doing so constitutes a violation of academic integrity and will result in disciplinary action if detected.

Tentative Outline

1. Week 1: Introduction to Mobile App Development, Flutter & Project Setup
2. Week 2: Dart Fundamentals
3. Week 3: Dart Advanced Topics
4. Week 4: Widgets
5. Week 5: Layouts, Navigation & Basic State Management
6. Week 6: Styling & Theming, Flutter Forms
7. Week 7: Lists
8. Week 8: JSON Parsing
9. Week 9: HTTP Requests
10. Week 10: Advanced State Management (Provider)
11. Week 11: Firebase, Authentication
12. Week 12: Caching
13. Week 13: Testing & Debugging
14. Week 14: App Deployment